

Education

Duke University - Junior

B.S.E. Mechanical Engineering, Aerospace Engineering Certificate; 3.97 GPA

Sep 2024 – May 2028

Courses: Thermodynamics, Fluid Dynamics, Control System, Junior Mechanical Design, Signals & Systems, Computer Architecture, Advanced Web Development, Constructal Theory & Design in Nature.

Employment

R&D Engineer Intern - Keshavarz Fluid Dynamics Group

May 2026 - Present

- High-speed photography of hydrodynamic instabilities in rotary atomization. Fuel injector design & combustion processes.

Design Engineer - Duke DesignHub

Dec 2025 - Present

- Provide engineering consultation & design services for Duke researchers & hospital clinicians to fulfill their design needs.

Machinist & Teaching Assistant - Duke Pratt Student Machine Shop

Sep 2025 - Present

- Support machining classwork, junior & senior design projects by providing machining expertise and shop safety oversight.

Mechanical Engineering Intern - GEA Group Vietnam - Liquid & Powder Technology

Apr 2024 - Aug 2024

- Designed & documented basic P&IDs; conducted on-site tank, piping & valves inspections & maintenance for F&B factories.
- Developed general EXCEL calculating tools for GEA's homogenizer flow & heat exchange systems.
- Projects involved: Lothamilk Yogurt Production Line Expansion & Ba Vi Milk Production Line Upgrade

Experience & Projects

Avionics & Separation Lead Engineer, Propulsion Engineer - Duke AERO

Jun 2025 - Present

- Re-engineered tri-blade airbrakes control system for 40% weight reduction and documented detailed system datasheet.
- Developed Monte Carlo framework for characterization of different control algorithms, including gain-scheduling, PID, and adaptive MPC for precise apogee targeting with airbrakes.
- Led redesign of canards system for 35% control authority increase & 20% reduction in manufacturing & integration time.
- Spearheaded novel composite designs: carbon-fiber overmolded canard blade on steel shaft & forged fiberglass access hatch.
- Manufactured 6.125" OD x 7.5" Aluminum control stack housing on VIWA CNC machine, optimizing cycle time by 50%.
- Designed first-gen bipropellant [liquid engine flight stack architecture](#), supporting cold-flow system characterization & testing.
- Full-cycle complete ownership of design, manufacture & characterization of new prop manifold & impinging injector design.

Avionics, Structures, Propulsion Design Engineer - Duke AERO

Sep 2024 - Jun 2025

- Led [FINSight V2](#): novel dual-layer PCB with embedded accelerometers & DAQ system to characterize rocket fin flutter.
- In charge of design & analysis of IP-67 rated, blast-proof [black box](#) housing & PCB, validated with ANSYS FEA.
- Led research on [canard](#)-tail fin aerodynamic coupling & control surface flutter for canard roll-stabilization system. Developed closed-loop PI ramping control process. Selected for podium presentation at IREC 2025 as an outstanding project.
- Conducted NOX & Kerosene injector simulations on ANSYS Fluent to validate injection pattern and pressure losses.
- Manufactured flight-critical [propulsion hardware](#), including 6" solid rocket motor closures, nozzle & APCP propellant grains.
- Designed modular tank & motor [test stands](#) & DAQ system for cold flow & static fire, withstanding over 35000Ns impulse.
- Assisted with development of LABView-based data logging & control system for ENO.

Technologies

Relevant Skills: ANSYS Structural & Fluent, GD&T, Tolerance Stackup, Precision Machining, Composite Layup & Molding, CAD/CAM (SolidWorks, Onshape, AutoCAD, Fusion360), PCB design (Altium, KiCAD, Eagle), DFM & AM Optimization, Aircraft Design Suite (Athena Vortex Lattice, XFOIL & XFLR5, OpenVSP, MachUp)

Programming, Control Languages & Frameworks: Python, C, C++, Java, MATLAB, MIPS Assembly, HTML, CSS, JS, TypeScript, Vue, React, Docker, ROS, LABView & NI-DAQmx